5 Sets of ICSE Class 10 Physics - PYQ Based Revision Material

Set 1

30 Most Important Numericals

1. **(2019)** Two bodies A and B have masses in the ratio 5:1 and their kinetic energies are in the ratio 125:9. Find the ratio of their velocities. [Ans: 5:3]

2. **(2024)** A door lock is opened by turning the lever (handle) of length 0.2 m. If the moment of force produced is 1 Nm, then the minimum force required is: [Ans: 5 N]

3. **(2022 Sem 2)** A metal piece of mass 420 g present at 80°C is dropped in 80 g of water present at 20°C in a calorimeter of mass 84 g. If the final temperature is 30°C, calculate specific heat capacity of metal. [Ans: 0.168 $Jg^{-1}°C^{-1}$]

4. **(2023)** Calculate the power spent by a crane while lifting a load of mass 2000 kg at velocity of 1.5 ms^{-1} (g = 10 ms^{-2}). [Ans: 30000 W]

5. **(2018)** A solid metal weighing 150 g melts at its melting point of 800°C by providing heat at 100 W for 4 minutes. Find specific latent heat. [Ans: 1.6×10^5 J/kg]

6. **(2024)** A radioactive nucleus containing 128 nucleons emits a β-particle. After β-emission, the number of nucleons present will be: [Ans: 128]

7. **(2016)** Calculate the mass of ice required to lower temperature of 300 g water at 40°C to water at 0°C. (L = 336 J/g, c = 4.2 J/g°C) [Ans: 150 g]

8. **(2020)** A pond appears to be 2.7 m deep. If refractive index of water is 4/3, find actual depth. [Ans: 3.6 m]

9. **(2022 Sem 2)** A wire of resistance 2Ω gets stretched to thrice its original length. Calculate new resistance. [Ans: 18Ω]

10. **(2015)** Rajan exerts a force of 150 N in pulling a cart at constant speed of 10 m/s. Calculate power. [Ans: 1500 W]

11. **(2019)** A pendulum has frequency 4 vibrations per second. Observer fires gun and hears echo after 6 vibrations. If v = 340 m/s, find distance to cliff. [Ans: 255 m]

12. **(2014)** 50 g of metal at 27°C requires 2400 J to attain 327°C. Calculate specific heat capacity. [Ans: 0.16 J/g°C]

13. **(2023)** A convex lens of focal length 10 cm is placed 60 cm from screen. Find object distance for real image. [Ans: 12 cm]

14. **(2017)** A pulley system with VR = 4 lifts load of 175 kgf through 15 m. Effort = 50 kgf. Calculate work done by effort. [Ans: 30000 J]

15. **(2024)** Sumit does 600 J work in 10 min and Amit does 300 J in 20 min. Calculate ratio of powers. [Ans: 4:1]

16. **(2013)** A force is applied on body of mass 20 kg moving with 40 ms^{-1}. Body attains 50 ms^{-1} in 2 s. Calculate work done. [Ans: 9000 J]

17. **(2021)** Calculate current drawn by appliance rated 110 W, 220 V when connected to 220 V supply. [Ans: 0.5 A]

18. **(2012)** A boy of mass 30 kg sits at 2 m from middle of see-saw. Where should 40 kg boy sit to balance? [Ans: 1.5 m]

19. **(2010)** A spring compressed by trolley of mass 0.5 kg, released to move at 2 ms^{-1}. Find potential energy of spring. [Ans: 1 J]

20. **(2022 Sem 1)** A body of mass 200 g falls freely from height 15 m. Find velocity with which it hits ground. [Ans: $10\sqrt{3}$ ms^{-1}]

21. **(2019)** 104 g water at 30°C in copper calorimeter of mass 42 g. Ice added, final temp 10°C. Find mass of ice. [Ans: 24 g]

22. **(2024)** A fuse is rated 5 A. Can it be used with geyser rated 1540 W, 220 V? [Ans: No, 7 A required]

23. **(2018)** Three resistors of 2Ω, 3Ω and 5Ω joined to get total resistance more than 7Ω. Find equivalent resistance. [Ans: 10Ω]

24. **(2023)** A coin kept inside water (μ = 4/3) appears raised by 3 mm. Find actual depth. [Ans: 12 mm]

25. **(2014)** Calculate change in kinetic energy if velocity is reduced to 1/3rd of initial. [Ans: 8/9 of initial KE]

26. **(2016)** A music system draws 400 mA at 12 V. Find resistance. When current drops to 320 mA, find voltage. [Ans: 30Ω, 9.6 V]

27. **(2020)** A nucleus ${}_{84}X^{202}$ emits α-particle followed by β-particle. Find a and b in final nucleus ${}_{a}Y^{b}$. [Ans: a=83, b=198]

28. **(2015)** A refrigerator converts 100 g water at 20°C to ice at -10°C in 35 minutes. Find average rate of heat extraction. [Ans: 34 W]

29. **(2013)** 50 g ice at 0°C added to 300 g liquid at 30°C. Find final temperature. (c liquid = 2.65 J/g°C) [Ans: 0°C (ice just melts)]

30. **(2022 Sem 1)** A woman draws water using fixed pulley. Mass of bucket = 10 kg. Force applied = 200 N. Find mechanical advantage. [Ans: 0.5]

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20 Formulas

1. $F = ma$ - Newton's Second Law
2. $W = F \times s \times \cos\theta$ - Work done
3. $P = W/t$ - Power
4. $KE = 1/2\ mv^2$ - Kinetic Energy
5. $PE = mgh$ - Potential Energy
6. $1/f = 1/v - 1/u$ - Lens Formula
7. $P = 1/f$ - Power of Lens (f in m)
8. $\mu = c/v$ = Real depth/Apparent depth - Refractive Index
9. $T = 2\pi\sqrt{(l/g)}$ - Time Period of Pendulum
10. $V = f\lambda$ - Wave Equation
11. $Q = mc\Delta T$ - Heat (Specific Heat)
12. $Q = mL$ - Heat (Latent Heat)
13. $V = IR$ - Ohm's Law
14. $P = VI = I^2R = V^2/R$ - Electrical Power
15. $E = Pt = VIt$ - Electrical Energy
16. $1/Rp = 1/R_1 + 1/R_2$ - Parallel Resistance
17. $Rs = R_1 + R_2$ - Series Resistance
18. MA = Load/Effort - Mechanical Advantage
19. VR = Effort distance/Load distance - Velocity Ratio
20. $\eta = (MA/VR) \times 100\%$ - Efficiency

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10 Diagrams

1. **(2019)** Ray diagram for concave mirror showing image formation when object is beyond C

2. **(2020)** Path of ray through rectangular glass slab when incident normally
3. **(2023)** Path of ray through equilateral prism at minimum deviation
4. **(2018)** Claw hammer as lever (Class I)
5. **(2022 Sem 2)** Magnetic compass near coil to determine polarity
6. **(2015)** Block and tackle system of pulleys with VR = 4
7. **(2017)** Ray diagram for concave lens with object at F
8. **(2021)** Dual control switch circuit for staircase wiring
9. **(2014)** Temperature-time graph for cooling of naphthalene
10. **(2016)** Circuit diagram showing series combination of resistors with ammeter and voltmeter

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10 Definitions

1. **Moment of Force (2014)**: The product of magnitude of force and the perpendicular distance of line of action of force from the axis of rotation.

2. **Power (2018)**: The rate at which work is done by a force. SI unit is watt (1 watt = 1 joule/second).

3. **Refractive Index (2009)**: The ratio of speed of light in vacuum to the speed of light in the medium.

4. **Specific Heat Capacity (2023)**: The amount of heat energy required to raise the temperature of 1 kg of a substance by 1°C.

5. **Critical Angle (2012)**: The angle of incidence in the denser medium for which the angle of refraction in the rarer medium is 90°.

6. **Mechanical Advantage (2015)**: The ratio of load lifted to the effort applied.

7. **Resonance (2015)**: When the frequency of forced vibrations equals the natural frequency of the body, the body vibrates with increased amplitude.

8. **Nuclear Fission (2019)**: The process in which a heavy nucleus splits into two lighter nuclei of comparable size with release of tremendous energy.

9. **Heat Capacity (2013)**: The amount of heat required to raise the temperature of a body by 1°C.

10. **Centre of Gravity (2014)**: The point where the entire weight of the body appears to act vertically downward.

Set 2

30 Most Important Numericals

1. **(2020)** A satellite revolves around Earth in circular orbit. What is work done by satellite at any instant? [Ans: Zero]

2. **(2018)** The power of a lens is -5 D. Find its focal length and name the lens. [Ans: -20 cm, concave lens]

3. **(2015)** A body of mass 50 kg has momentum 3000 kg ms^{-1}. Calculate kinetic energy. [Ans: 90000 J]

4. **(2023)** A geyser is rated 240 W - 220 V. Explain the meaning. [Ans: At 220 V, consumes 240 J/s]

5. **(2019)** A man standing between two cliffs produces sound and hears echoes at 3 s and 4 s. Find distance between cliffs. (v = 330 m/s) [Ans: 1155 m]

6. **(2022 Sem 1)** A stone tied at end of string is whirled in horizontal circle with uniform speed. Name the force required. [Ans: Centripetal force]

7. **(2014)** 50 g of ice at 0°C is added to 300 g of liquid at 30°C. Find final temperature. [Ans: 0°C]

8. **(2016)** Calculate heat produced in 20Ω resistor carrying 2.5 A for 5 minutes. [Ans: 37500 J]

9. **(2024)** A force F moves load from A to C. Displacement is: [Ans: 5 m]

10. **(2013)** A body of mass 0.2 kg falls from 10 m to 6 m above ground. Find loss in PE. (g = 10 ms^{-2}) [Ans: 8 J]

11. **(2021)** A piece of ice of mass 60 g is dropped into 140 g water at 50°C. Find final temperature. [Ans: 11°C]

12. **(2012)** Calculate equivalent resistance between points A and B. [Ans: 5Ω]

13. **(2017)** An electric bulb of 300Ω draws 0.4 A. Calculate power and potential difference. [Ans: 48 W, 120 V]

14. **(2020)** Two metallic blocks P and Q have masses in ratio 2:1, same heat supplied, same temperature rise. Compare specific heat capacities. [Ans: $S_P = ½ S_Q$]

15. **(2015)** A lens forms erect, magnified virtual image. Name the lens. Focal length 25 cm. Calculate power. [Ans: Convex lens, +4 D]

16. **(2023)** A car of mass 120 kg moves at 18 km/h and accelerates to 54 km/h in 5 s. Calculate work done. [Ans: 12000 J]

17. **(2019)** A magnet kept at centre of two coils A and B is moved. Which galvanometer shows more deflection and why? [Ans: B, more turns]

18. **(2010)** A body of mass 1.5 kg is dropped from height 12 m. Find force acting during fall. [Ans: 14.7 N]

19. **(2022 Sem 2)** An appliance rated 440 W, 220 V is connected across 220 V. Calculate maximum current. [Ans: 2 A]

20. **(2016)** A crane 'A' lifts load in 5 s, crane 'B' does same work in 2 s. Compare power. [Ans: $P^A : P^B = 2 : 5$]

21. **(2024)** A ray of light is incident normally on face of equilateral prism. Ray gets totally reflected. Find deviation. [Ans: 60°]

22. **(2018)** A half metre rod is pivoted at centre with weights 20 gf at 6 cm and 12 gf at 10 cm. Is rod in equilibrium? [Ans: Yes]

23. **(2013)** Calculate the quantity of heat produced in coil of 75Ω if 2 A passed for 2 minutes. [Ans: 36000 J]

24. **(2014)** Find equivalent resistance between points A and B. [Ans: 7Ω]

25. **(2023)** A ray of light is incident at 45° on equilateral prism. Name the phenomenon when it enters and emerges. [Ans: Refraction, dispersion]

26. **(2015)** A person standing between two cliffs 480 m from nearest cliff shouts. Hears first echo after 3 s and second after 2 s later. Find speed of sound. [Ans: 320 m/s]

27. **(2021)** Calculate the mass of ice needed to cool 150 g water in calorimeter of mass 50 g from 32°C to 5°C. [Ans: 30.6 g]

28. **(2012)** A body of mass 5 kg moving with 10 ms^{-1}. Find ratio of initial to final KE if mass doubled and velocity halved. [Ans: 2:1]

29. **(2019)** A nuclear fusion reaction: $^{3}He_{2} + ^{2}H_{1} \rightarrow ^{4}He_{2}$ + _____ Complete the reaction. [Ans: $^{1}H_{1}$ + energy]

30. **(2020)** Calculate total resistance across AB: 3Ω, 6Ω, 5Ω. [Ans: 7Ω]

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20 Formulas

1. $p = mv$ - Linear Momentum
2. $F = \Delta p/\Delta t$ - Force from momentum change
3. $v = u + at$ - First equation of motion
4. $s = ut + \frac{1}{2}at^2$ - Second equation of motion
5. $v^2 = u^2 + 2as$ - Third equation of motion
6. $\mu = \sin i/\sin r$ - Snell's Law
7. $\sin ic = 1/\mu$ - Critical angle relation
8. $f = R/2$ - Focal length of spherical mirror
9. $m = v/u = hi/ho$ - Magnification
10. $T = 1/f$ - Time period
11. Heat capacity = m × c - Relation between heat capacity and specific heat
12. $P = W/t = Fv$ - Power (alternate form)
13. $1 \text{ kWh} = 3.6 \times 10^6 \text{ J}$ - Relation between kWh and joule
14. $R = \rho\, l/A$ - Resistance in terms of resistivity
15. MA = Effort arm/Load arm - Mechanical advantage of lever
16. $\varepsilon = V + Ir$ - EMF and terminal voltage
17. Load × Load arm = Effort × Effort arm - Principle of moments
18. $Q = I^2Rt$ - Heat produced in resistor
19. Specific resistance = R × A/l - Resistivity
20. $F = BIl \sin\theta$ - Force on current-carrying conductor in magnetic field

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10 Diagrams

1. **(2020)** Path of monochromatic light through equilateral glass prism
2. **(2019)** Ray diagram for concave lens showing virtual image
3. **(2023)** Block and tackle pulley system with VR = 4
4. **(2018)** Right-angled isosceles prism used to make inverted erect image
5. **(2022 Sem 1)** Balanced metre scale with weights at ends

6. **(2015)** Diagram showing refraction through glass slab with lateral displacement
7. **(2021)** Magnetic field lines around a bar magnet with neutral point
8. **(2016)** V-I graph for ohmic resistor
9. **(2014)** Single movable pulley diagram
10. **(2017)** Ray diagram for convex lens when object between F and 2F

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10 Definitions

1. **Velocity Ratio (2009)**: The ratio of velocity of effort to the velocity of load.

2. **Total Internal Reflection (2010)**: The phenomenon when a ray of light travelling from denser to rarer medium is reflected back into the denser medium at the interface.

3. **Latent Heat of Fusion (2013)**: The amount of heat required to convert 1 kg of solid at its melting point into liquid at the same temperature.

4. **Nuclear Fusion (2018)**: The process in which two or more smaller nuclei combine to form a bigger nucleus with release of large amount of energy.

5. **Power of Lens (2014)**: The measure of deviation produced by a lens in the path of rays refracted through it.

6. **Centripetal Force (2015)**: The force acting on a body moving in a circular path, directed towards the centre of the circle.

7. **Echo (2009)**: The repetition of sound caused by reflection of sound waves from a distant surface.

8. **Ohm's Law (2016)**: The current flowing through a conductor is directly proportional to the potential difference across its ends, provided physical conditions remain constant.

9. **Specific Latent Heat (2012)**: The amount of heat required to change the state of 1 kg of substance without any change in temperature.

10. **Radioactivity (2008)**: The spontaneous emission of α, β and γ radiations from the nucleus of an atom.

Set 3

30 Most Important Numericals

1. **(2019)** A body of mass 10 kg is kept at height 5 m. It is allowed to fall. Find total mechanical energy at height 2 m. [Ans: 500 J]

2. **(2022 Sem 2)** In given circuit, 12Ω and 8Ω are in parallel with 2Ω in series. Calculate external resistance and current I_2 if I_1 = 0.4 A. [Ans: 6.8Ω, 0.6 A]

3. **(2023)** A metre scale of weight 50 gf balanced at 40 cm mark without any weight. If cut at centre, which part weighs more? [Ans: 0-50 cm part]

4. **(2016)** A boy weighing 40 kgf climbs 30 steps each 20 cm high in 4 min. Girl weighing 30 kgf does same in 3 min. Compare work done and power. [Ans: Work 4:3, Power 1:1]

5. **(2024)** A non-uniform beam of weight 120 N pivoted at one end. Calculate force F to keep beam in equilibrium. [Ans: 30 N]

6. **(2015)** A radar sends signal to aircraft at 30 km away and receives back after 2×10^{-4} s. Find speed. [Ans: 3×10^{8} m/s]

7. **(2021)** 30 g of ice at 0°C is used to bring down temperature of certain mass of water at 70°C to 20°C. Find mass of water. [Ans: 60 g]

8. **(2013)** An object is placed at distance 24 cm in front of convex lens of focal length 8 cm. Find image distance and magnification. [Ans: 12 cm, -0.5]

9. **(2012)** A moving body weighing 400 N possesses 500 J of KE. Calculate velocity. (g = 10 ms^{-2}) [Ans: 5 ms^{-1}]

10. **(2020)** A crane lifts load of 1000 kg to height 10 m in 20 s. Calculate power. [Ans: 5000 W]

11. **(2018)** A person standing at sea shore. Observer on ship anchored between cliff and shore fires gun. Person hears two sounds at 2 s and 3 s. Find distance between observer and shore. [Ans: 320 m]

12. **(2014)** A ray of light passes from water to air. How does speed change? [Ans: Increases]

13. **(2023)** Calculate total heat required to melt 200 g ice at 0°C to water at 100°C. [Ans: 151200 J]

14. **(2015)** An object is placed at distance 24 cm in front of convex lens of focal length 8 cm. Nature of image? [Ans: Real, inverted, diminished]

15. **(2019)** A pendulum has frequency 4 vibrations per second. Time for 6 vibrations = ? [Ans: 1.5 s]

16. **(2022 Sem 1)** A body of mass 200 g falls from 15 m. Find potential energy at 10 m above ground. [Ans: 20 J]

17. **(2013)** A force is applied on body of mass 20 kg moving with 40 ms^{-1}. Body attains 50 ms^{-1} in 2 s. Calculate acceleration. [Ans: 5 ms^{-2}]

18. **(2016)** A boy uses blue light to find refractive index. Then uses red light. Will refractive index be same? [Ans: Different]

19. **(2024)** A radioactive element P emits α-particle. Position of Q in periodic table? [Ans: Two groups to left]

20. **(2012)** A hot solid of mass 60 g at 100°C placed in 150 g water at 20°C. Final temp 25°C. Find specific heat of solid. [Ans: 0.6 J/g°C]

21. **(2015)** A block and tackle system has VR = 5. Rohan exerts pull of 150 kgf. Maximum load if efficiency = 75%. [Ans: 562.5 kgf]

22. **(2021)** Calculate the amount of heat absorbed by 200 g of paraffin wax to melt completely. (L = 146 Jg^{-1}) [Ans: 29200 J]

23. **(2010)** A boy weighs 360 N on earth. What would be his weight on moon? [Ans: 60 N]

24. **(2023)** A ray of light travels normal to given surface. Angle of refraction? [Ans: 0°]

25. **(2019)** A uniform metre scale is in equilibrium with 40 gf at 5 cm and unknown weight at 30 cm. Find weight of scale. [Ans: 50 gf]

26. **(2014)** Calculate current in each resistor when 4Ω and 6Ω are in parallel drawing 0.5 A from cell. [Ans: 0.3 A, 0.2 A]

27. **(2020)** A nucleus ${}_{84}X^{202}$ emits α-particle. Find new nucleus. [Ans: ${}_{82}Y^{198}$]

28. **(2016)** A sound wave travelling in water has wavelength 0.4 m. Is it audible in air? (v in water = 1400 ms^{-1}) [Ans: Yes, 3500 Hz]

29. **(2022 Sem 2)** A copper conductor placed over stretched wires connected to DC supply. Determine polarity for upward force. [Ans: N at A, S at B]

30. **(2013)** A piece of ice of mass 40 g is dropped into 200 g water at 50°C. Find final temperature. [Ans: 28.3°C]

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20 Formulas

1 kgf = 9.8 N - Relation between kgf and newton

Moment = F × d - Moment of force

Work = mgh - Work against gravity

Power = Work/Time - Power definition

np/ns = Vp/Vs = Is/Ip - Transformer relation

μ = 1/sin ic - Refractive index from critical angle

Lateral displacement = t × sin(i-r)/cos r - Lateral shift in glass slab

$s = (v^2 - u^2)/2a$ - Displacement from velocity change

$F = G(m_1m_2)/r^2$ - Newton's law of gravitation

$g = GM/R^2$ - Acceleration due to gravity

Efficiency = (Output work/Input work) × 100% - Efficiency

VR = Number of strands supporting load in movable block - VR of pulley system

1 cal = 4.186 J - Relation between cal and joule

$1\ eV = 1.6 \times 10^{-19}$ J - Electron volt to joule

$P = V^2/R$ - Power from voltage and resistance

$F = GMm/r^2$ - Gravitational force

W = QV = VIt - Electrical work

1 HP = 746 W - Horse power to watt

$P_1/P_2 = (A_1/A_2)^2$ - Loudness vs amplitude

$1\ \mu = 10^{-6}$ - Micro prefix

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10 Diagrams

1. **(2020)** Complete path of ray AB through equilateral prism PQR
2. **(2018)** Diagram showing scattering of light (Rayleigh scattering)
3. **(2022 Sem 2)** Three-pin socket with colour coding of wires

4. **(2015)** Lemon crusher showing direction of load and effort
5. **(2023)** Ray diagram for convex lens with object at 2F
6. **(2019)** Dual control switch circuit for bulb
7. **(2021)** Magnetic field lines of Earth using compass needle
8. **(2016)** V-I graph for series and parallel combination
9. **(2014)** Diagram showing forces on ball hanging by string
10. **(2017)** Ray diagram for concave mirror with object beyond C

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10 Definitions

1. **Scalar Quantity (IX Syllabus)**: A physical quantity that has only magnitude and no direction. Example: mass, distance, speed.

2. **Vector Quantity (IX Syllabus)**: A physical quantity that has both magnitude and direction. Example: force, velocity, displacement.

3. **Acceleration due to Gravity (IX Syllabus)**: The uniform acceleration produced in a freely falling body due to the gravitational force of the earth.

4. **Buoyancy (IX Syllabus)**: The upward force experienced by a body when immersed in a fluid.

5. **Relative Density (IX Syllabus)**: The ratio of density of a substance to the density of water at 4°C.

6. **Work Function (2008)**: The minimum amount of energy required to eject an electron from a metal surface.

7. **Isobars (2018)**: Atoms of different elements having the same mass number but different atomic numbers.

8. **Isotopes (2009)**: Atoms of the same element having the same atomic number but different mass numbers.

9. **Free Vibrations (2018)**: Vibrations of a body with constant amplitude and frequency in the absence of any external force.

10. **Damped Vibrations (2017)**: Vibrations whose amplitude gradually decreases with time due to the presence of some damping force.

Set 4

30 Most Important Numericals

1. **(2024)** A radioactive element is placed in evacuated chamber. Rate of radioactive decay will: [Ans: Remain unchanged]

2. **(2019)** A body of mass 10 kg is kept at height 5 m. Find kinetic energy just before hitting ground. [Ans: 500 J]

3. **(2022 Sem 1)** A boy standing in front of wall produces two whistles per second. Echo coincides with sound. Find distance to wall if v = 320 m/s. [Ans: 80 m]

4. **(2016)** A stone of mass 64 g thrown upward with 20 ms^{-1} from ground. Calculate potential energy at maximum height. [Ans: 12.8 J]

5. **(2023)** Calculate power of engine of car mass 120 kg accelerating from 18 km/h to 54 km/h in 5 s. [Ans: 2400 W]

6. **(2015)** A nut is opened by wrench of length 20 cm with force 2 N. Find moment of force. [Ans: 0.4 Nm]

7. **(2020)** A pond appears 2.7 m deep. If μ = 4/3, find actual depth. [Ans: 3.6 m]

8. **(2014)** 40 g of water at 60°C poured into vessel containing 50 g water at 20°C. Final temp 30°C. Find thermal capacity of vessel. [Ans: 294 J/°C]

9. **(2018)** An electric iron rated 220 V, 2 kW used for 2 h daily for a week. Find cost at ₹4.25 per kWh. [Ans: ₹119]

10. **(2013)** A body of mass 0.2 kg falls from 10 m to 6 m. Find loss in potential energy. [Ans: 8 J]

11. **(2022 Sem 2)** In circuit with 12Ω and 8Ω in parallel, I_1 = 0.4 A. Find I_2 and total current I. [Ans: 0.6 A, 1 A]

12. **(2012)** A ray of light incident at angle i passes through equilateral prism with refracted ray parallel to base. Relation between i and e? [Ans: i = e]

13. **(2019)** A piece of ice of mass 40 g dropped into 200 g water at 50°C. Find final temperature. [Ans: 28.3°C]

14. **(2024)** A fuse is rated 5 A. Can it be used with geyser rated 1540 W, 220 V? [Ans: No]

15. **(2015)** A person standing between two cliffs 480 m from nearest cliff shouts. Hears first echo after 3 s. Find distance of other cliff. [Ans: 720 m]

16. **(2016)** A music system draws 400 mA at 12 V. Find resistance. [Ans: 30Ω]

17. **(2023)** A coin inside water (μ = 4/3) appears raised by 3 mm. Find actual depth. [Ans: 12 mm]

18. **(2013)** Calculate equivalent resistance between points A and B. [Ans: 5Ω]

19. **(2021)** A metal piece of mass 5 g has thermal capacity 2.5 JK^{-1}. If mass tripled, find specific heat capacity. [Ans: 0.5 $Jg^{-1}K^{-1}$]

20. **(2014)** A ray of light passes from water to air. How does speed change? [Ans: Increases]

21. **(2018)** A solid metal weighing 150 g melts at 800°C with heat rate 100 W for 4 min. Find specific latent heat. [Ans: 1.6×10^5 J/kg]

22. **(2012)** A body of mass 5 kg moving with 10 ms^{-1}. Find KE. [Ans: 250 J]

23. **(2020)** A nucleus ${}_{84}X^{202}$ emits α-particle followed by β-particle. Find final nucleus. [Ans: ${}_{83}Y^{198}$]

24. **(2015)** Calculate speed of sound if person hears echo after 3.5 s and after moving 84 m closer hears after 3 s. [Ans: 336 m/s]

25. **(2019)** A uniform metre scale balanced with 40 gf at 5 cm and unknown weight at 30 cm. Find weight of scale. [Ans: 50 gf]

26. **(2013)** A force applied on body of mass 20 kg changes velocity from 40 to 50 ms^{-1} in 2 s. Find work done. [Ans: 9000 J]

27. **(2016)** A crane lifts load in 5 s, another does same work in 2 s. Compare power. [Ans: 2:5]

28. **(2024)** A ray of light incident normally on face of equilateral prism gets totally reflected. Find deviation. [Ans: 60°]

29. **(2014)** 50 g of ice at 0°C added to 300 g liquid at 30°C. Find final temperature. [Ans: 0°C]

30. **(2012)** A hot solid of mass 60 g at 100°C placed in 150 g water at 20°C. Final temp 25°C. Find specific heat of solid. [Ans: 0.6 J/g°C]

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20 Formulas

1. $s = (u+v)/2 \times t$ - Average velocity × time
2. $a = (v-u)/t$ - Acceleration
3. $F = GMm/R^2$ - Gravitational force
4. $g = GM/R^2$ - g from G
5. $W = mg$ - Weight
6. $p = h\rho g$ - Pressure due to liquid column
7. Upthrust = $V\rho g$ - Buoyant force
8. RD = Density of substance/Density of water - Relative density
9. $\mu = c/v$ - Refractive index
10. $V = f\lambda$ - Wave velocity
11. $T = 2d/v$ - Time for echo
12. $Q = mc\Delta T + mL$ - Heat with phase change
13. $\varepsilon = I(R+r)$ - EMF with internal resistance
14. $R = V/I$ - Resistance from V-I
15. $P = I^2R$ - Power from current
16. $E = (P \times t)/1000$ kWh - Electrical energy in kWh
17. Cost = Units × Rate - Electricity bill
18. $Ep/Es = Np/Ns = Is/Ip$ - Transformer
19. $A = Z + N$ - Mass number relation
20. 1 Å $= 10^{-10}$ m - Angstrom to metre

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10 Diagrams

1. **(2020)** Graph between displacement and time for free vibrations
2. **(2018)** Circuit diagram with ammeter and voltmeter to verify Ohm's law
3. **(2023)** Path of parallel rays A (Orange) and B (Blue) through air-glass boundary
4. **(2015)** Pulley system with VR = 3
5. **(2022 Sem 1)** Magnetic compass near current-carrying conductor
6. **(2019)** Ray diagram for convex mirror
7. **(2021)** Heating curve of water showing phase changes
8. **(2016)** Diagram showing forces on forearm as lever
9. **(2014)** Staircase wiring diagram
10. **(2017)** AC generator labelled diagram

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10 Definitions

1. **Inertia (IX Syllabus)**: The natural tendency of an object to resist any change in its state of rest or of uniform motion.

2. **Force (IX Syllabus)**: An external agency that changes or tends to change the state of rest or uniform motion of a body.

3. **Pressure (IX Syllabus)**: The thrust (force) acting perpendicularly on a unit area of a surface.

4. **Archimedes' Principle (IX Syllabus)**: When a body is immersed wholly or partially in a fluid, it experiences an upward force equal to the weight of the fluid displaced.

5. **Principle of Floatation (IX Syllabus)**: A body floats in a liquid if the weight of the liquid displaced by the immersed part of the body is equal to the weight of the body.

6. **Thermionic Emission (2015)**: The emission of electrons from a metal surface when heat energy is imparted to it.

7. **Nuclear Waste (2013)**: The radioactive by-product material produced from nuclear reactors or nuclear fission processes.

8. **Background Radiation (2012)**: The low-level radiation constantly present in the environment from natural and artificial sources.

9. **Forced Vibrations (2012)**: Vibrations of a body under the influence of an external periodic force.

10. **Resistivity (2017)**: The resistance offered by a conductor of unit length and unit cross-sectional area. It depends only on the material of the conductor.

Set 5

30 Most Important Numericals

1. **(2024)** The graph shows heat energy supplied vs temperature change. The slope gives: [Ans: Heat capacity]

2. **(2019)** A body of mass 10 kg is kept at height 5 m. Find total mechanical energy. [Ans: 500 J]

3. **(2022 Sem 1)** A woman draws water using fixed pulley. Mass = 10 kg. Force = 200 N. Find MA. [Ans: 0.5]

4. **(2018)** Temperature of 170 g water at 50°C lowered to 5°C by adding ice. Find mass of ice. [Ans: 90 g]

5. **(2023)** A convex lens of focal length 10 cm placed 60 cm from screen. Find object distance. [Ans: 12 cm]

6. **(2016)** A boy weighing 40 kgf climbs 30 steps each 20 cm high in 4 min. Find power in watt. [Ans: 10 W]

7. **(2020)** A crane lifts load of 1000 kg to height 10 m in 20 s. Calculate power. [Ans: 5000 W]

8. **(2015)** A radar sends signal to aircraft at 45 km away. Speed = 3×10^{8} ms^{-1}. Find time for signal to return. [Ans: 3×10^{-4} s]

9. **(2014)** A hot solid of mass 60 g at 100°C placed in 150 g water at 20°C. Final temp 25°C. Find specific heat of solid. [Ans: 0.6 J/g°C]

10. **(2022 Sem 2)** A wire of resistance 2Ω stretched to thrice its length. Find new resistance. [Ans: 18Ω]

11. **(2013)** A body of mass 0.2 kg falls from 10 m to 6 m. Find loss in PE. [Ans: 8 J]

12. **(2024)** A fuse is rated 5 A. Geyser rated 1540 W, 220 V. Can it be used? [Ans: No]

13. **(2012)** A moving body weighing 400 N possesses 500 J of KE. Find velocity. [Ans: 5 ms^{-1}]

14. **(2019)** A pendulum has frequency 4 vibrations per second. Time for 6 vibrations? [Ans: 1.5 s]

15. **(2016)** A music system draws 400 mA at 12 V. Find voltage when current drops to 320 mA. [Ans: 9.6 V]

16. **(2023)** Calculate power spent by crane lifting 2000 kg at 1.5 ms^{-1}. [Ans: 30000 W]

17. **(2021)** 30 g ice at 0°C used to cool water from 70°C to 20°C. Find mass of water. [Ans: 60 g]

18. **(2015)** A nut opened by wrench of length 20 cm with force 2 N. Find moment. [Ans: 0.4 Nm]

19. **(2020)** A nucleus $_{84}X^{202}$ emits α-particle. Find new nucleus. [Ans: $_{82}Y^{198}$]

20. **(2014)** 40 g of water at 60°C poured into vessel with 50 g water at 20°C. Final temp 30°C. Find thermal capacity of vessel. [Ans: 294 J/°C]

21. **(2018)** An electric iron rated 220 V, 2 kW used for 2 h daily for a week. Find cost at ₹4.25/kWh. [Ans: ₹119]

22. **(2022 Sem 2)** A metal piece of mass 420 g at 80°C dropped in 80 g water at 20°C in calorimeter of mass 84 g. Final temp 30°C. Find specific heat of metal. [Ans: 0.168 $Jg^{-1}°C^{-1}$]

23. **(2013)** Calculate equivalent resistance between A and B. [Ans: 5Ω]

24. **(2012)** A hot solid of mass 60 g at 100°C placed in 150 g water at 20°C. Final temp 25°C. Find specific heat of solid. [Ans: 0.6 J/g°C]

25. **(2024)** Sumit does 600 J work in 10 min and Amit does 300 J in 20 min. Ratio of powers? [Ans: 4:1]

26. **(2015)** An object placed at distance 24 cm in front of convex lens of focal length 8 cm. Find image distance. [Ans: 12 cm]

27. **(2019)** A uniform metre scale balanced with 40 gf at 5 cm and unknown weight at 30 cm. Find weight of scale. [Ans: 50 gf]

28. **(2014)** Calculate change in KE if velocity reduced to 1/3rd. [Ans: 8/9 of initial]

29. **(2016)** A boy uses blue and red light to find refractive index. Will it be same? [Ans: Different]

30. **(2023)** A coin inside water (μ = 4/3) appears raised by 3 mm. Find actual depth. [Ans: 12 mm]

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20 Formulas

1. 1 Nm = 10^7 dyne cm - Moment of force unit conversion
2. 1 J = 10^7 erg - Work unit conversion
3. $F = mv^2/r$ - Centripetal force

4. $T = 2\pi\sqrt{(l/g)}$ - Simple pendulum
5. LC = 1 MSD - 1 VSD - Least count of Vernier
6. LC = Pitch/No. of divisions on circular scale - Least count of screw gauge
7. $p = F/A$ - Pressure
8. $FB = V\rho g$ - Buoyant force
9. $\rho_1/\rho_2 = V_2/V_1$ - Floatation relation
10. 1 cal = 4.186 J - Calorie to joule
11. $C = Q/\Delta T$ - Heat capacity
12. $c = Q/m\Delta T$ - Specific heat capacity
13. $L = Q/m$ - Latent heat
14. $m_1c_1\Delta T_1 = m_2c_2\Delta T_2$ - Principle of calorimetry
15. $n = \sin i/\sin r$ - Refractive index
16. $f = uv/(u+v)$ - Focal length from u and v
17. $P = 1/f$ (f in m) - Power of lens
18. $\delta = i + e - A$ - Angle of deviation in prism
19. $N_p/N_s = V_p/V_s = I_s/I_p$ - Transformer turns ratio
20. ${}^{A}_{2}X \rightarrow {}^{A-4}_{2}Y + {}^{4}_{2}He$ - Alpha decay

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10 Diagrams

1. **(2020)** Path of ray through rectangular glass slab with normal incidence
2. **(2018)** Diagram showing U-shaped soft iron electromagnet
3. **(2023)** Ray diagram for convex lens with object at 2F
4. **(2015)** Block and tackle system of pulleys with VR = 5
5. **(2022 Sem 2)** Magnetic field lines around current-carrying loop
6. **(2019)** Ray diagram for concave mirror with object at F
7. **(2021)** Circuit showing household wiring with ring main
8. **(2016)** Graph showing variation of volume of water with temperature (0-10°C)
9. **(2014)** Diagram of a hot cathode ray tube
10. **(2017)** Ray diagram for total internal reflection in prism

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10 Definitions

1. **Uniform Circular Motion (X Syllabus)**: Motion of a body along a circular path with constant speed. Although speed is constant, velocity changes due to change in direction, hence it is accelerated motion.

2. **Centrifugal Force (X Syllabus)**: The outward force experienced by a body moving in a circular path when observed from a rotating frame of reference.

3. **Mechanical Advantage (X Syllabus)**: The ratio of load lifted to the effort applied in a machine.

4. **Velocity Ratio (X Syllabus)**: The ratio of distance moved by the effort to the distance moved by the load in the same time interval.

5. **Efficiency (X Syllabus)**: The ratio of useful work output to the total work input, expressed as percentage.

6. **Critical Angle (2012)**: The angle of incidence in the denser medium for which the angle of refraction in the rarer medium is 90°.

7. **Dispersion of Light (2012)**: The phenomenon of splitting of white light into its constituent colours when passing through a prism.

8. **Scattering of Light (2018)**: The phenomenon of bending of light at the corners of small objects and apertures, causing light to spread out.

9. **Nuclear Fission (2018)**: The process in which a heavy nucleus splits into two lighter nuclei of comparable size with release of tremendous energy.

10. **Nuclear Fusion (2018)**: The process in which two or more smaller nuclei combine to form a bigger nucleus with release of large amount of energy.